



CARTAN

AI-Native Arthroplasty

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Problem

The knee implant that **patients prefer most** is the knee implant that **surgeons use least**

Problem

Modern TKA sacrifices the ACL leading to **unnatural kinematics, loss of function,** and **decreased patient satisfaction.**

Bicruciate-retaining (**BCR**) TKA preserves the ACL but is **too difficult** for most surgeons.

80%  of TKA patients report their knees feel **“artificial”**¹

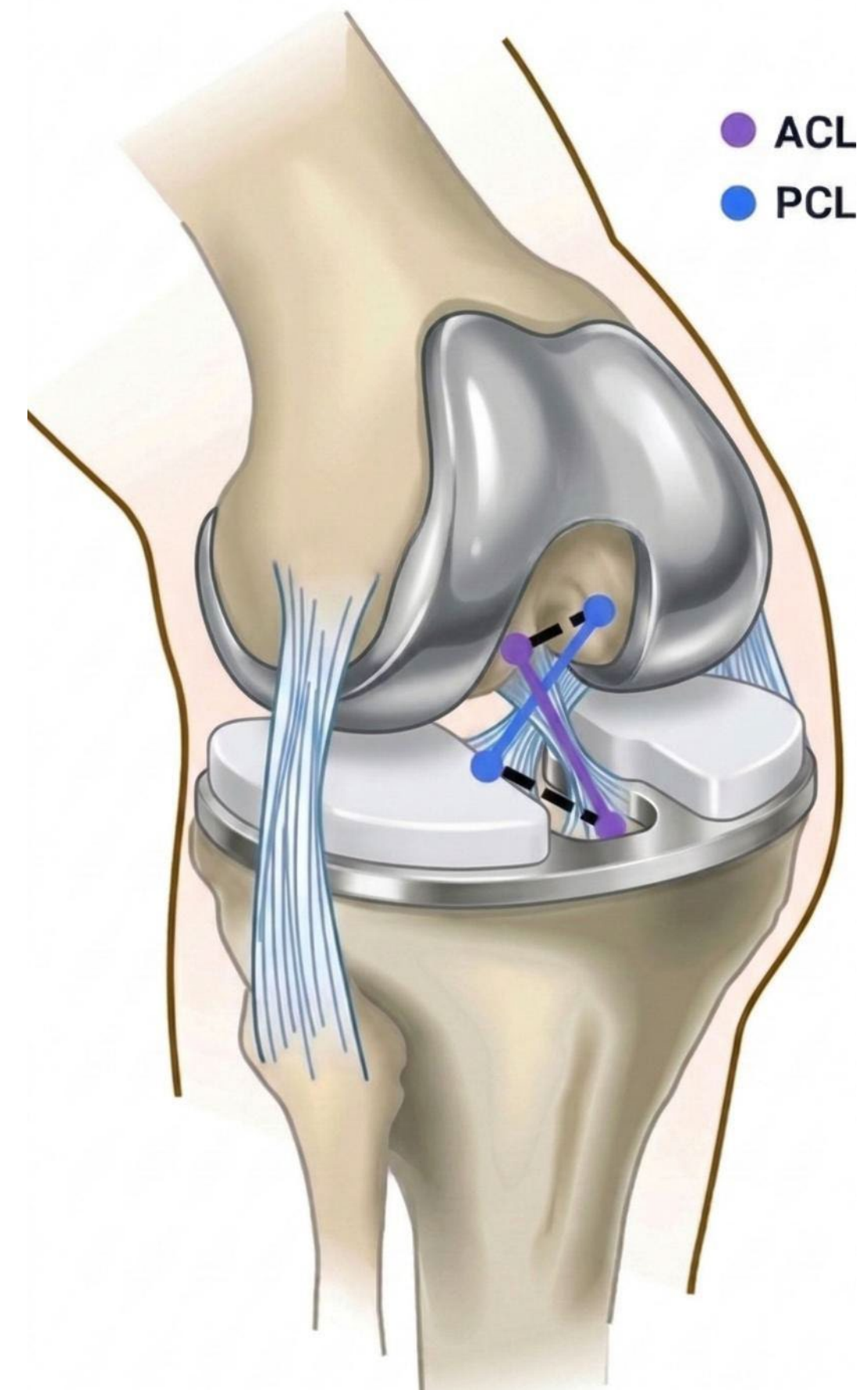
BCR Benefits

Bicruciate-Retaining (BCR) TKA

Preserving the cruciate ligaments maintains the natural **four-bar linkage** that drives natural knee kinematics and **cannot be reproduced** by an implant.

Patients **Prefer BCR** Implants

Among bilateral TKA patients, **89% preferred BCR** over CS (ACL & PCL sacrificed), and **74% preferred BCR** over CR (ACL sacrificed), noting that the BCR felt **more “normal” and “stable.”**²



BCR Barrier to Adoption

BCR is so mechanically complex that it fundamentally **requires** AI to be performed reliably.

“BCR TKA implies **major technical challenges** and specific complications resulting from ACL retention and **difficult tissue balancing**.”³

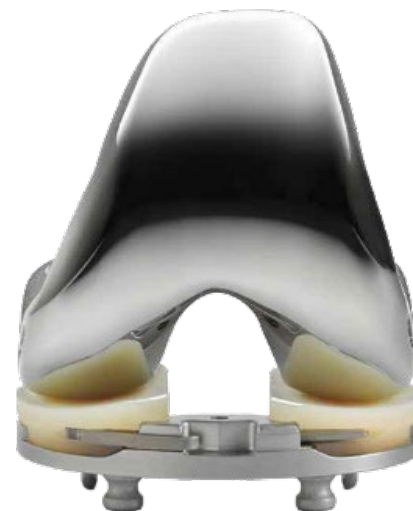
“**Optimal ligament balancing** for an ACL- and PCL-preserved knee has **yet to be defined**.”⁴

First Generation BCR (1972-2016)

Great short- and long-term results from expert surgeons. Extreme technical difficulty and a high learning curve prevented mainstream adoption.⁵



Fig. 1, Pritchett⁵



Zimmer Vanguard XP (2016-2019)

Non-anatomic implant design and surgical technique created kinematic conflict between the cruciate ligaments and implant geometry, driving high complication and revision rates.

S&N Journey II XR (2018-2025)

Anatomic implant design and **robotic implementation** led to use by a small but enthusiastic group of expert surgeons. **Robotics did not drive adoption**, and S&N withdrew the XR during a recent portfolio rationalization.



3. Sabatini L, Barberis L, Camazzola D, Centola M, Capella M, Bistolfi A, Schiraldi M, Massè A. Bicruciate-retaining total knee arthroplasty: What's new? *World J Orthop.* 2021; 12(10): 732-742
4. De Windt TS, Van Laarhoven SN, Van Hellemond GG. Bicruciate-Retaining Total Knee Arthroplasty: State of the Art for the Younger and Active Population? A Systematic Review and Future Prospective. *Applied Sciences.* 2022; 12(21):10721.
5. Pritchett JW. Bicruciate-retaining Total Knee Replacement Provides Satisfactory Function and Implant Survivorship at 23 Years. *Clin Orthop Relat Res.* 2015 Jul;473(7):2327-33.

Solution: AI-Native Arthroplasty

Surgical Data Streams

Absorbing the Surgeon Cognitive Load

Cartan's intelligence platform acts as an **ambient operating layer**.

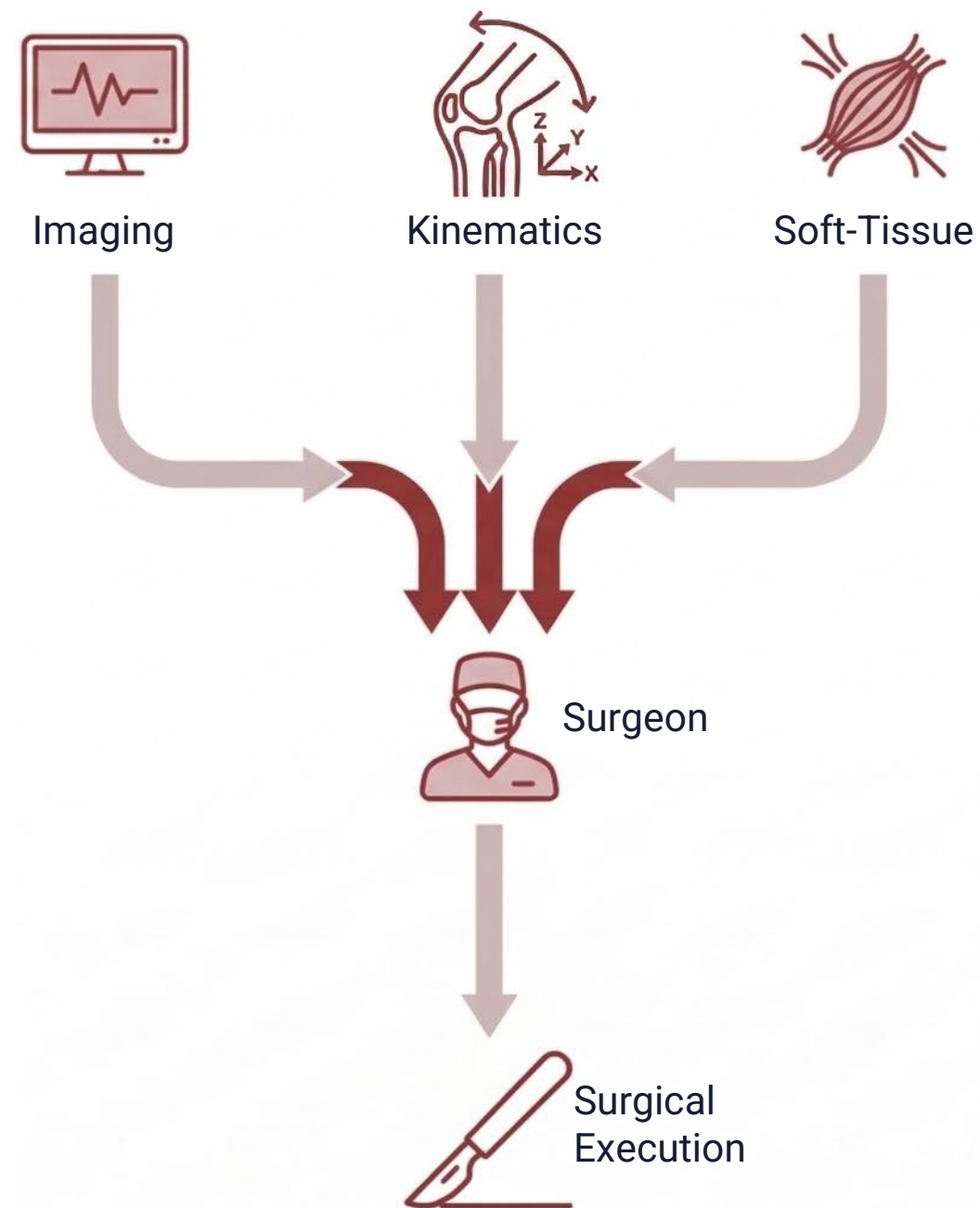
Unlocking Biological Preservation

AI-Native Arthroplasty makes **BCR** TKA **accessible**, **scalable**, and **highly reproducible**.

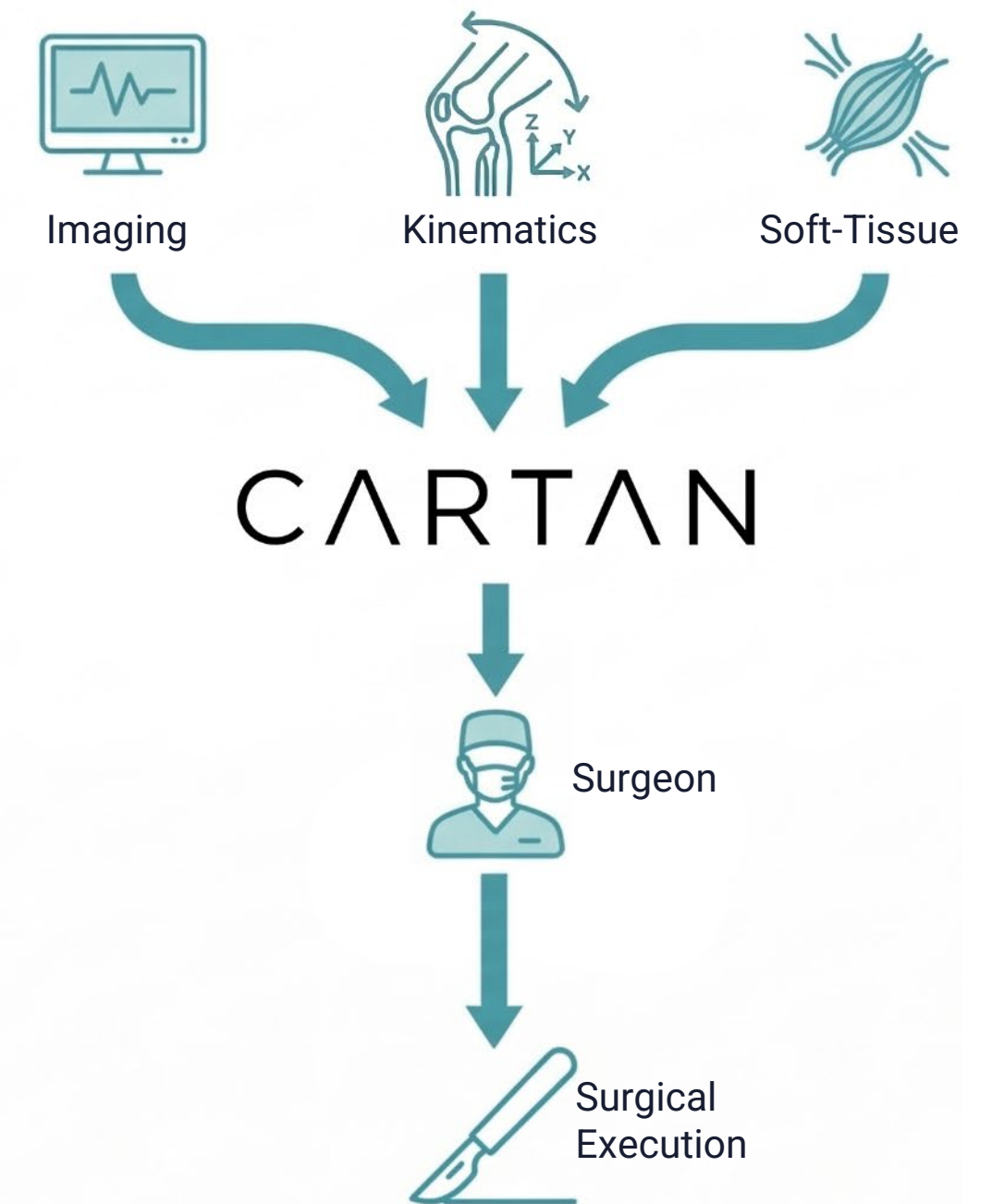
Surgeon-In-The-Loop

The surgeon maintains **full visibility and control** throughout the surgical procedure.

Traditional Arthroplasty



AI-Native Arthroplasty



AI-Native Arthroplasty: Components

Surgical AI Stack

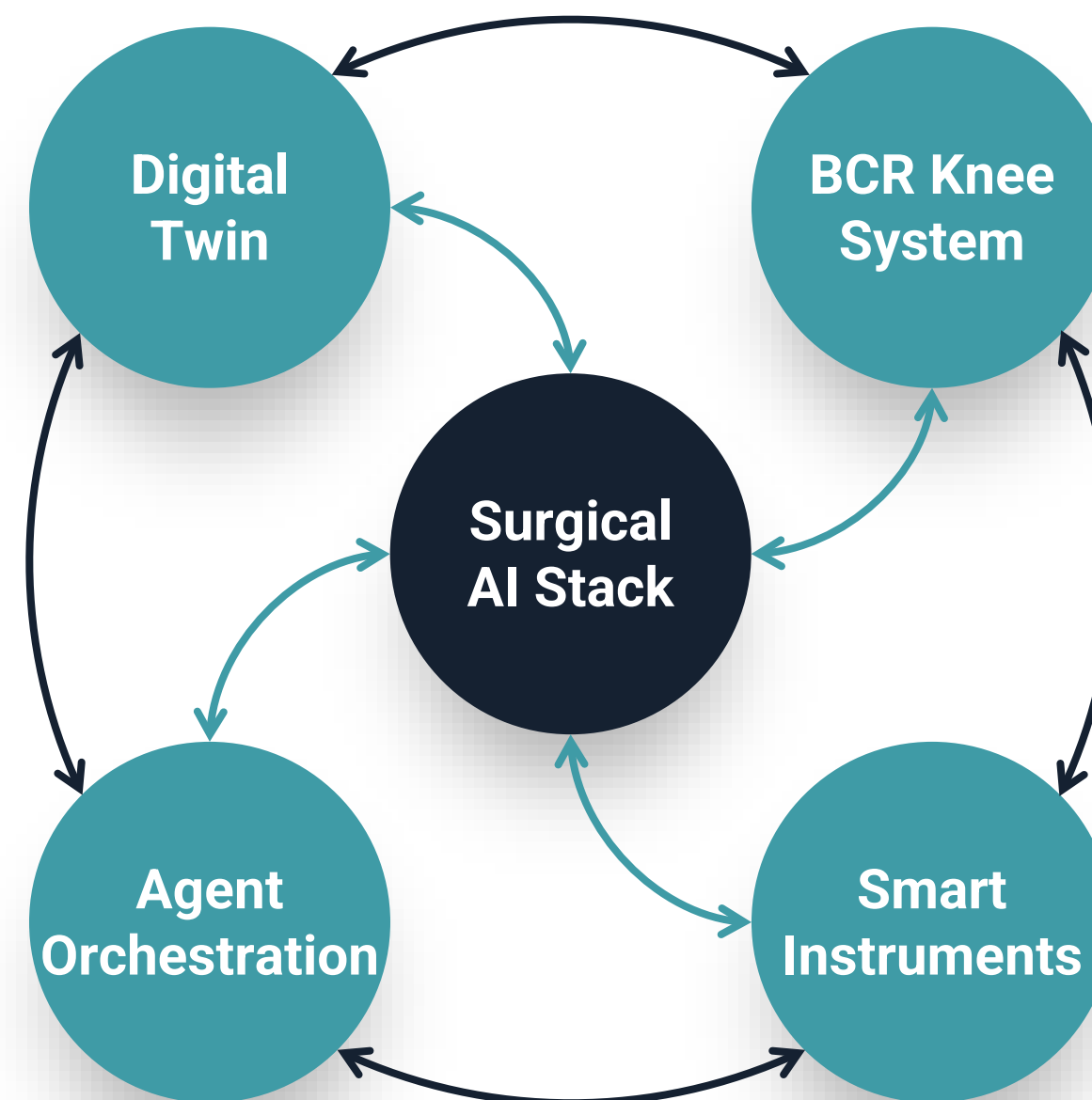
Modular, multimodal AI stack, purpose-built to learn **TKA surgical workflows** and advanced **physical, spatial** and **temporal** reasoning

Digital Twin

Multimodal physics-based digital twins are used to **simulate** and **visualize** kinematics and complication risks

Agent Orchestration

Surgical Support: pre-, intra-, post-op automation to direct planning, surgical, and post-op workflows.
Continuous Learning: **aggregate** and **deidentify** case data for **continuous learning**.



Knee Systems

Niche Entry Point: **cementless BCR** implant with modular inserts to fine tune medial and lateral tibial slope

Family and Market Expansion: rapid expansion to offer Cruciate Retaining (**CR**) and Cruciate Sacrificing (**CS**) implants.

Smart Instruments

Intuitive tools to quantify **key physiological parameters** in real time for maximal observability and surgeon control

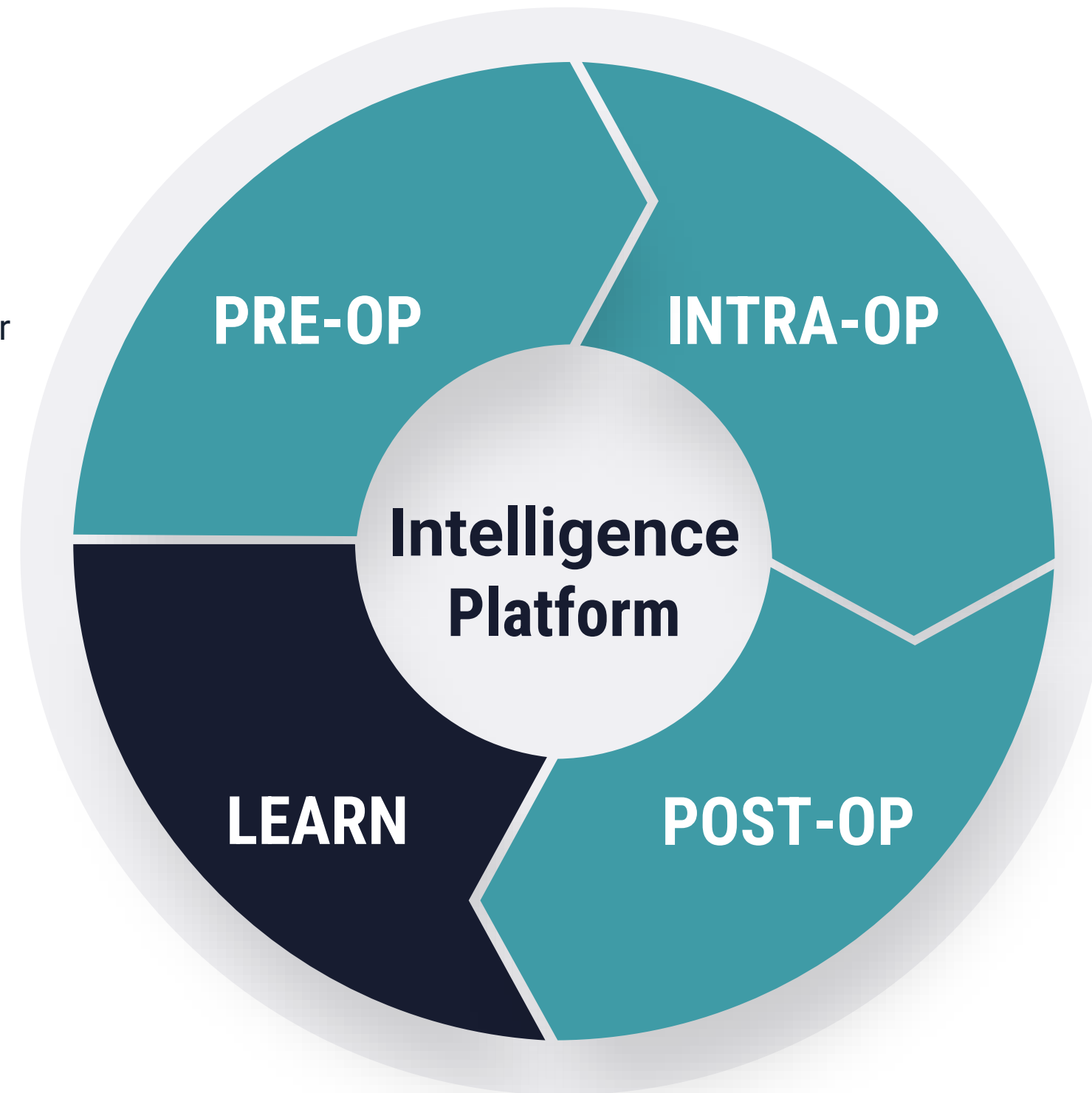
AI-Native Arthroplasty: Workflow

Plan + Simulate

- **identify** optimal implant placement
- dynamic **predictive kinematics** and wear
- **quantify** complication risk
- dynamic pre- and post-op **kinematic visualization**

Collect + Train

- **compile** deidentified datasets
- continuously **monitor** performance
- **improve** predictive capabilities



Sense + Adapt + Guide

- **measure** and **verify** key parameters
- **fine-tune** the digital twin and surgical plan in real-time
- precise adjustments of smart cutting guide or robotic system

Monitor + Validate

- validate **implant position** and **kinematics**
- monitor **risk vectors**
- **identify** complications early

Why Now?

Rapid **technological advancements** and **favorable market conditions** create the ideal opportunity to introduce and scale Cartan's AI-native BCR procedure.

Advances in Physical AI

- **PhysicsNeMo** framework enables real-time, physics-informed digital twins trained on simulation data, simplifying AI infrastructure development.
- **Cosmos** now includes surgical-specific foundation models that learn tissue deformation and tool interaction directly from procedure data, dramatically reducing the cost of synthetic training data generation.
- Advances in **neural operators**, **graph neural networks**, and **diffusion models** allow complex biomechanical systems to be simulated at interactive speed.

Ebbing Ortho-MedTech Innovation

- Orthopedic robotics became a decade-long market-share battle that **eroded margins**.
- Robot commoditization has shifted incumbent prioritization to **operational efficiency** over **innovation**.
- The current dry spell of innovation is an opportunity to gain **market visibility** and **market share**.

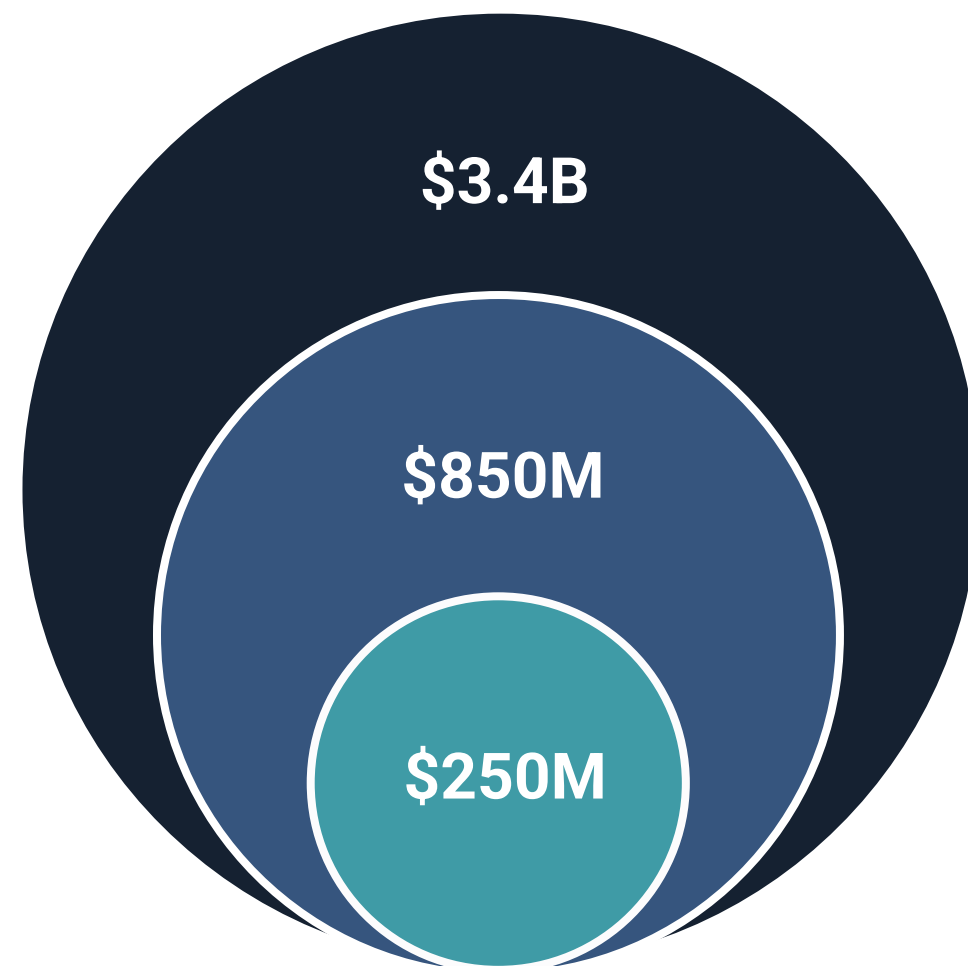
Unmet Surgeon Demand

- **64%** of surgeons believe there is a **need for BCR** TKA.⁶
- Smith & Nephew's withdraw of the Journey II XR in late 2025 marks the first time in over a decade that **no BCR is available in the US**.

6. De Faoite D, Ries C, Foster M, Boese CK. Indications for bi-cruciate retaining total knee replacement: An international survey of 346 knee surgeons. PLoS One. 2020 Jun 15;15(6)

Market Size

BCR is a **technical proving ground** and **strategic entry point** to the broader **\$3.4B** primary TKA market



TAM

US Primary TKA Market

Currently valued at **\$3.4B** and growing at **5.6% CAGR**; shifts towards ASCs and younger demographics both heavily **favor BCR adoption**.

SAM

BCR Segmentation

On average, surgeons report that **20% of Primary TKA** patients are candidates for BCR;⁶ a premium implant with enabling technology can support a **25% price premium**.

SOM

Filling the Unmet Surgeon Demand

With **explicit surgeon demand** and **no direct BCR competition**, Cartan will **scale quickly** to fill the palpable BCR vacuum.

6. De Faoite D, Ries C, Foster M, Boese CK. Indications for bi-cruciate retaining total knee replacement: An international survey of 346 knee surgeons. PLoS One. 2020 Jun 15;15(6)

Incumbent Competition

Entrenched Yet Vulnerable Oligopoly

The **Big Four** control **85%** of knee and hip implant sales,⁷ but they are struggling to keep up with technical advances.

No BCR is Currently Sold in the US

BCR is a strategic entry point to the broader TKA market, with innate surgeon demand and **no direct competition**.

BCR is the Ideal AI Proving Ground

The prohibitively **steep learning curve** and **severe cognitive demands** of BCR make it the perfect technical benchmark for **AI-Native Arthroplasty**.

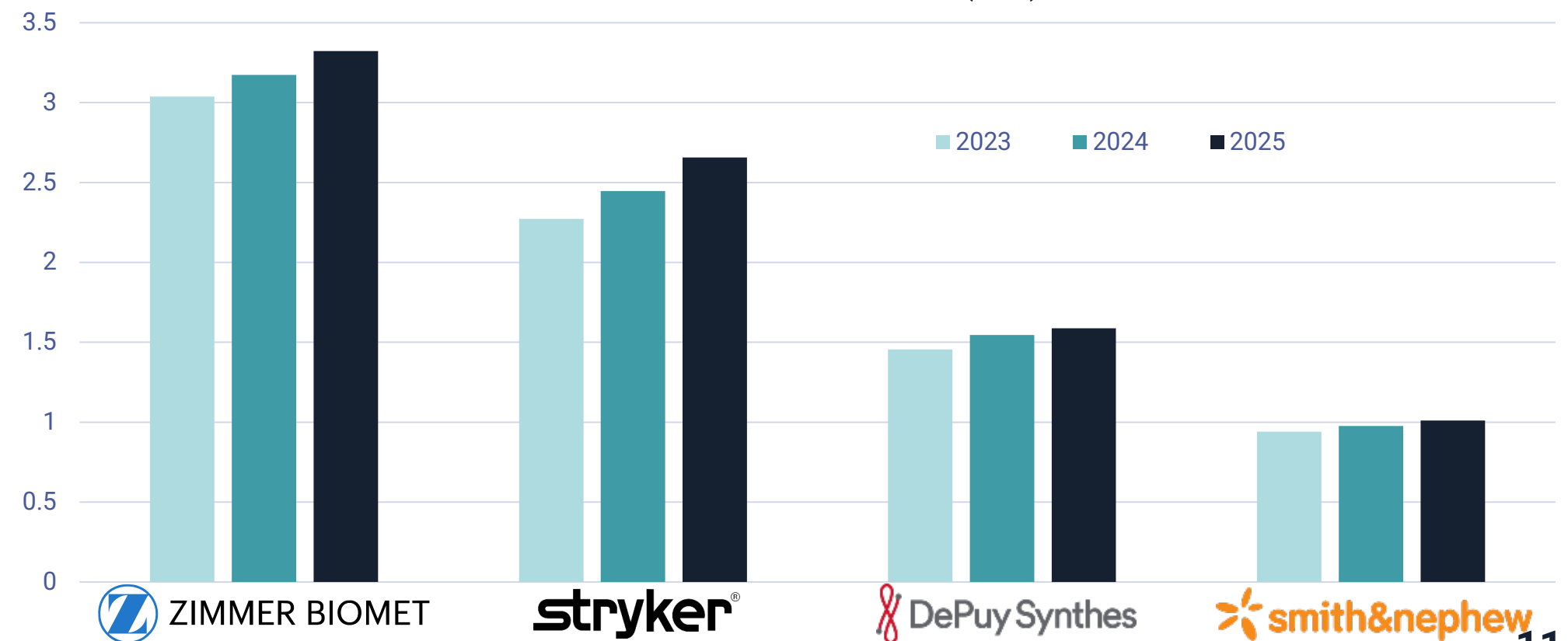


Struggled with robotics and each **lost market share** over the past decade



Gained **6.5%** market share over the last decade, largely attributable to its **acquisition of MAKO**⁸

Global Knee Revenues (\$B)



7. Tsai J. Orthopedics Market 2025 Update. Gabelli. <https://gabelli.com/research/orthopedics-market-2025-update/>. 2025 Apr 22.

8. Orthoworld. Who's Winning in Joint Replacement? Market Share, Robotics & 2025 Outlook. YouTube. <https://youtu.be/ySIsfidUtTs?t=48>. 2026 Feb 12.

The AI Arms Race in Orthopedics

“This is the future. There’s no question **AI will be proliferated across all of Stryker.**”⁹

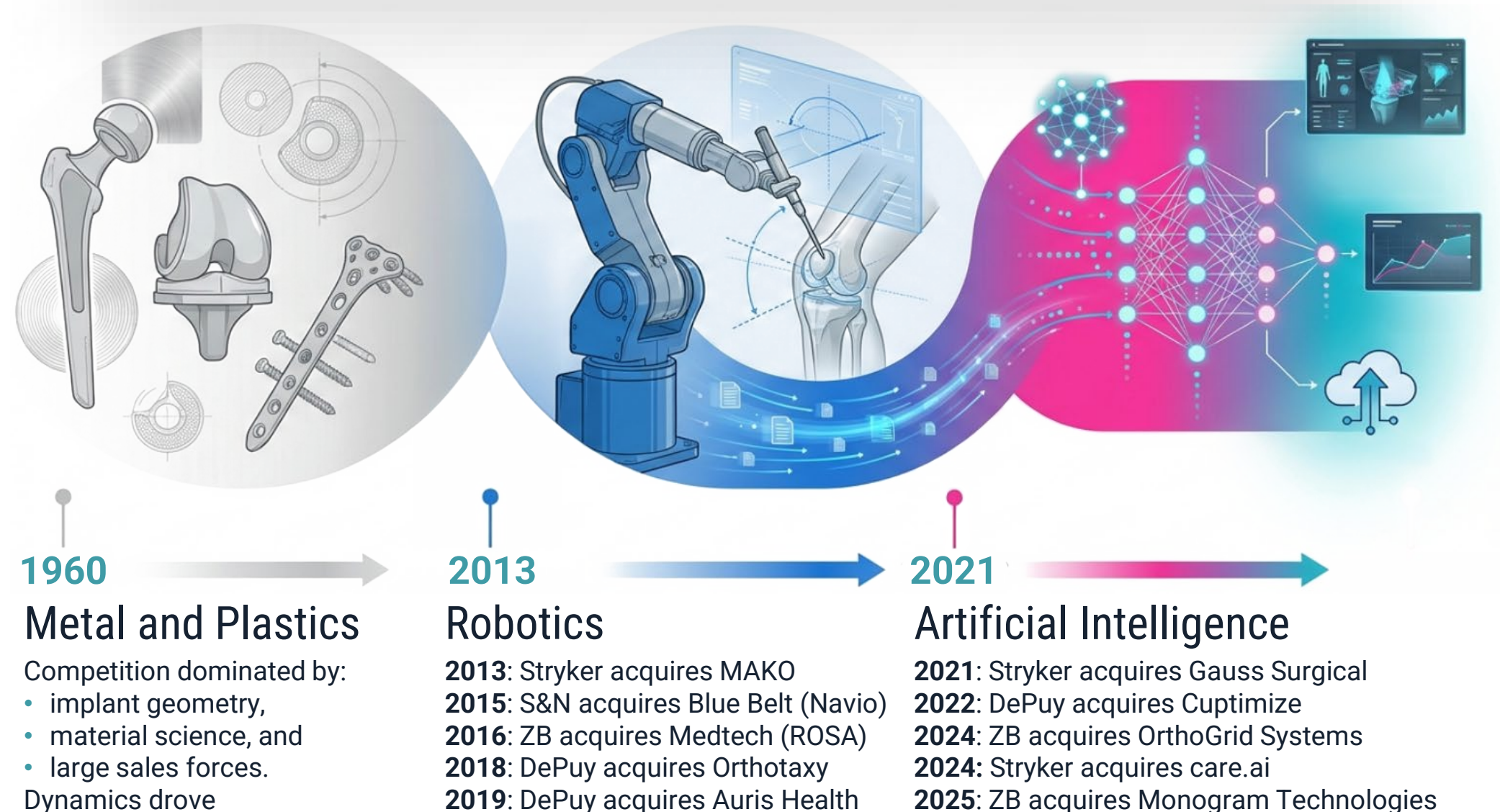
Kevin Lobo, CEO, Stryker

“We’re on this **new phase of digital and AI** which can really improve science. Bringing better science to the procedures and surgeries that are today a little bit of art.

Providing **operationalized workflows, automated procedures** and **safer surgery**. All of this is being enabled by this new digital and tech evolution and revolution, which **will affect all of the procedures.**”¹⁰

Kevin Lobo, CEO, Stryker

Incumbents are **actively seeking** orthopedic AI acquisitions.
Cartan is the **MAKO** of Orthopedic AI.



9. Kevin Lobo. Artificial intelligence will be proliferated across all of Stryker: CEO Kevin Lobo. Fox Business. <https://www.foxbusiness.com/video/6319894156112>. 2023 Feb 6.

10. The MedTech Conference hosted by AdvaMed. The Power of Digital Innovation and AI at #MTC25 with Stryker Chair & CEO, Kevin Lobo. YouTube. <https://www.youtube.com/watch?v=nOvGYlkc2ww>. 2025 Oct 7.

Regulatory Pathway

Implant Hardware

BCR Implant

Device Class: Class II

Path: Traditional 510(k)

Classification:

Knee joint patellofemoral tibial metal/polymer porous-coated uncemented prosthesis (21 CFR 888.3565)

Knee joint patellofemoral tibial polymer/metal/polymer semi-constrained cemented prosthesis (21 CFR 888.3560)

Product Code: JWH, MBH

Primary Predicate (BCR):

Journey II XR Knee System (K141471)

Secondary Predicate (Cementless):

Triathlon Total Knee System (K172326)

Note: Subject to Special Controls¹¹

Software and Instruments

Intelligence Platform

Device Class: Class II Software as a Medical Device (SaMD)

Path: Traditional 510(k)

Classification:

Orthopedic Stereotaxic Instrument (21 CFR 882.4560)

Product Code: OLO

Primary Predicate (Automated Planning):

Rosa Knee System (K251314)

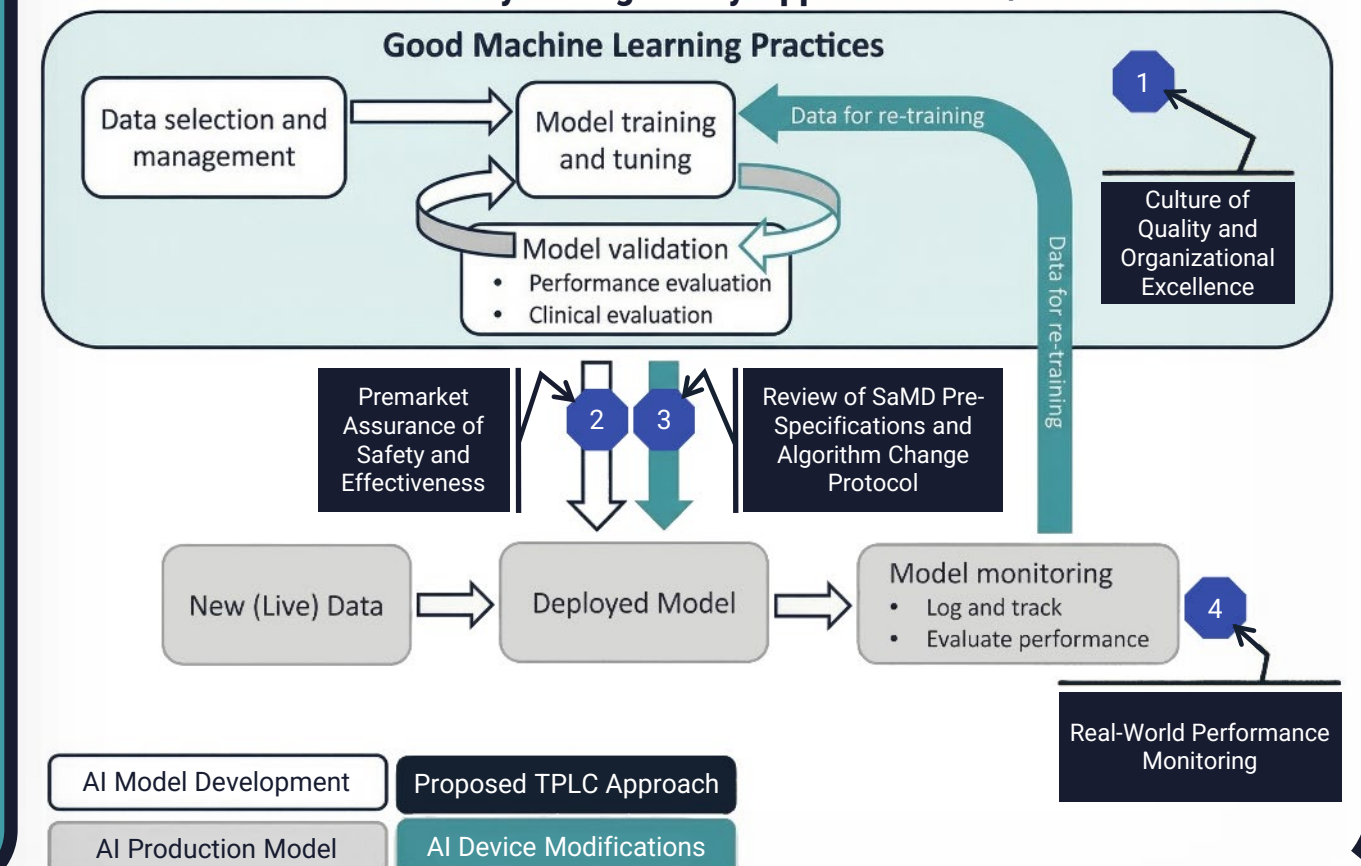
Secondary Predicate (Inertial Measurements):

KneeAlign 2 System (K172326)

To deploy **continuous learning** with a Class 2 SaMD Cartan will leverage the FDA's Predetermined Change Control Plan (**PCCP**)¹² framework during premarket review, including submission of:

- **SaMD Pre-Specifications (SPS):** rigorously defining the scope of modifications to performance, inputs, and/or intended use, and
- **Algorithm Change Protocol (ACP):** setting forth specific protocols for effecting the modifications delineated in the SPS.

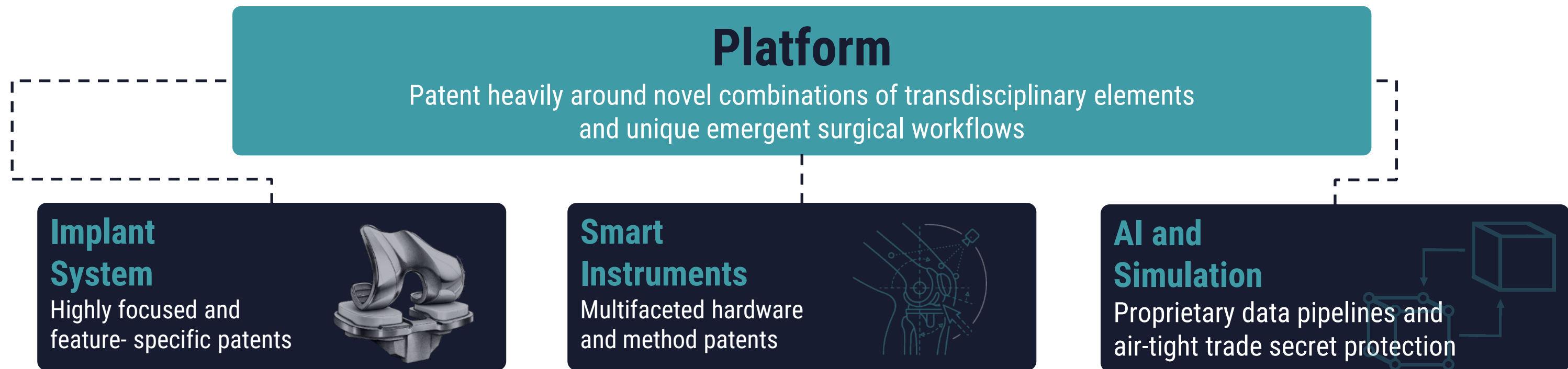
FDA's Total Product Lifecycle Regulatory Approach for AI/ML-Based SaMD



11. Knee Joint Patellofemoral and Femoral Tibial Metal/Polymer Porous-Coated Uncemented Prostheses - Class II Special Controls Guidance Document for Industry and FDA. <https://www.fda.gov/medical-devices/guidance-documents-medical-devices-and-radiation-emitting-products/knee-joint-patellofemoral-and-femoral-tibial-metal-polymer-porous-coated-uncemented-prostheses>. 2003 Jan 16

12. Proposed Regulatory Framework for Modifications to Artificial Intelligence/Machine Learning (AI/ML)-Based Software as a Medical Device (SaMD). <https://www.fda.gov/files/medical%20devices/published/US-FDA-Artificial-Intelligence-and-Machine-Learning-Discussion-Paper.pdf>. 2019 Apr 2

IP Dominance*



Proven IP Playbook

Led by Cartan's CEO, former Director of IP at **MAKO Surgical**, who designed and executed the patent portfolio and patent infringement litigation that:

- compelled **Stanmore's exit** from orthopedic robotics through a settlement requiring sale of its robotics assets and IP to MAKO, along with a commitment not to reenter the market
- preserved commercial separation between MAKO and **BlueBelt's** encroaching handheld platform
- contributed to the strategic pressure that culminated in **Stryker's** acquisition of MAKO at a \$1.7B valuation

Aggressive Strategy

- pursuing an **aggressive, multi-layered** IP strategy including prolific patent filings and trade-secret protection around proprietary AI/simulation data pipelines
- anticipating **100+ patent filings** anticipated by the end of 2026

Team: Executives



Jonathan Trousdale

CEO 

Domain Expertise: 15+ years building orthopedic medtech companies.

Moat Builder: History of building and enforcing dominant IP and technical moats.

Track Record of Exits: Planned and executed strategies that drove landmark acquisitions, including **MAKO Surgical (\$1.65B)** and **OrthoSensor**.

Chicago Booth MBA | Vanderbilt JD | Berkeley BS Eng



Russ Lee

COO 

Operational Excellence: Deployed sophisticated automation-first operations and led large-scale professional teams.

Scale Specialist: Proven track record of managing multi-billion-dollar P/Ls.

Executive Versatility: Former COO at **Cementify**, Chief Underwriting Officer at **Farmers**, and Head of Prod. Mgmt. at **Kemper**.

Chicago Booth MBA | OU MA Econ | West Point BS Mgt



Diego Berta

CBDO 

Ecosystem Architect: 15+ years creating and scaling frontier technology ecosystems.

Organizational Catalyst: Drives collaboration and alignment across engineering, product, legal, operations, and commercial teams.

Industry Connectivity: High-value network across frontier tech cultivated during senior leadership tenures at **MAKO Surgical**, **Magic Leap**, and **Peritas AI**.

Pace MBA | Brooklyn College BS Finance

Team: Directors and Advisors



Jon Greenwald

Director 

Orthopedic Authority: Decades of experience building and commercializing orthopedic implants, surgical navigation, and intraoperative sensors.

Commercialization Engine: Recognized innovator with a consistent track record of advancing complex surgical technologies from concept to market.

Industry Veteran: Currently CEO of [Caira Surgical](#), with previous senior leadership tenures at [4WEB Medical](#) and [OrthoSensor](#).

Columbia MFA | UCLA Anderson Board Director Cert. | Skidmore College BA



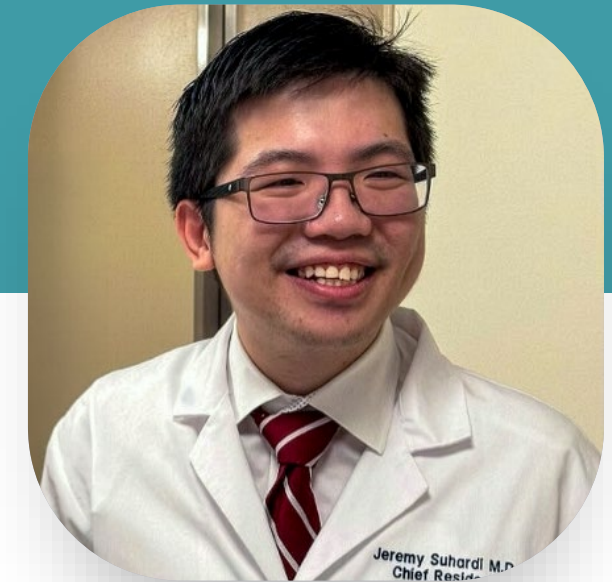
Morteza Meftah, MD

Director 

Clinical Authority: Board-certified orthopedic surgeon specializing in hip and knee reconstruction, complex total joint replacement, and robotic/computer-assisted surgery.

Innovation Leader: Drives surgical advancement as Co-Director of Robotic & Computer-Assisted Joint Replacement Surgery and Chair of the Center for Orthopedic Innovation at NYU Langone.

Academic & Industry Pedigree: Clinical Associate Professor at [NYU Langone](#), Director and former President of [ISTA](#), and active member of [AAOS](#) and [AAHKS](#).



Jeremy Suhardi, MD, PhD

Advisor 

Clinician-Scientist: Orthopedic surgeon specializing in adult reconstruction and the advancement of next-generation joint replacement therapies.

Translational Pioneer: Operates in both medical and deep tech domains, focusing on translational orthopedic research and the application of AI in surgical environments.

Institutional Excellence: Incoming Harvard/Brigham clinician-scientist and current [NYU Langone](#) fellow, backed by advanced degrees from [Caltech](#), [Harvard Medical School](#), and a PhD in Biomedical Engineering from [MIT](#).

Seed-Round Fundraising

Currently Raising \$4M

Post-Money SAFE

\$25M Valuation Cap

No Discount



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